

Homework[4]: Orbital Dynamic

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1. LINKS TO CODE

Here is a link to the code file [HW4 N body.ipynb](#). And all simulation gif files can be found under [gifs](#)

2. QUESTION 1: EARTH ORBITING THE SUN

2.1. Part a:

We integrated the motion for 3 orbital period (3 years) with timestep 0.01 years. The result is available at: [Earth animation GIF](#). We can see that Leapfrog produces a better approximation for Earth's orbit.

2.2. Part b:

We plot the speed of Earth as a function of time, as shown in [Earth velocity GIF](#) and in Figure 1.

We can see that the leapfrog method maintains a stable oscillation around the earth's orbit, while Euler consistently overshoot the orbit and decrease speed.

2.3. Part c:

We plot the total specific energy, ϵ , with the equation

$$\epsilon = \frac{1}{2}v^2 - \frac{GM_{\text{sun}}}{r}$$

. The result is shown in [Earth energy GIF](#)

2.4. Part d:

Comparing the result of Euler and Leapfrog, the leapfrog orbit is stable, and it conserves energy more accurately.

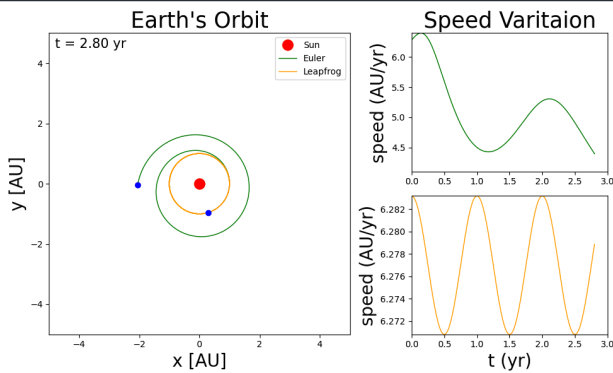


Figure 1. Earth's orbital speed over time

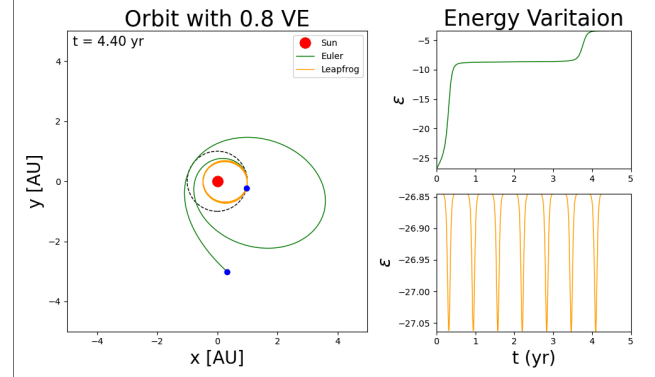


Figure 2. Earth's orbital specific energy over time, with initial orbit speed of $0.8 V_e$.

3. QUESTION 2: SMALLER INITIAL VELOCITY

Now we are considering a orbit with an initial velocity smaller than the orbital velocity.

3.1. Part a:

Here is the trajectory plot over 5 periods: [Earth speed 0.8 \$V_e\$ GIF](#), We observes similar results: a stable but elliptical orbit for Leapfrog, and a much more erratic orbit for Euler.

3.2. Part b:

Here is the energy plot over 5 periods: [Earth energy 0.8 \$V_e\$ GIF](#) and in Figure 2, We observes similar results: Leapfrog preserves the energy much better than Euler.

4. QUESTION 3 VOYAGER

Now we are considering a simple system, in which that Jupiter is influenced by only the Sun, and Voyager probe which is influenced by the Sun and Jupiter.

4.1. Part a:

With some trial and errors, if we launch Jupiter at 94.5 degree from Earth, and Voyager at an initial speed of 42.1 km/s, we can get a gravitational slingshot by Jupiter, as shown in [Jupiter Slingshot GIF](#) and in Figure 3. One failed attempt is shown in [Jupiter bounce GIF](#) where the probe is launched back to Earth instead!

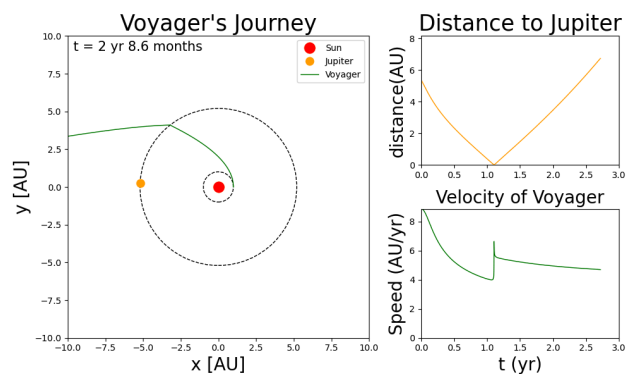


Figure 3. Gravitational Slingshot of Voyager by Jupiter.

4.2. Part b:

The closest distance is 0.004583 AU, with a speed of 6.058896 AU/yr, and 1.10 year after launch. The speed skyrocketed at the approach due to the close interaction with Jupiter's gravitational field.

5. QUESTION 4: SATURN?

As we consider Saturn as another boost source, it is much harder as Saturn is much lighter than Jupiter. One attempt is shown in [Jupiter and Saturn Slingshot GIF](#), where the closest Voyager-Saturn distance is 0.004616 AU at $t = 2.200$ yr.

ACKNOWLEDGMENTS

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